NAME:______________________ MATRIC NUMBER:______________

SCHOOL OF MATERIALS SCIENCE & ENGINEERING

CONTINUOUS ASSESSMENT SEMESTER 2 2021-2022

MS1016 – THERMODYNAMICS OF MATERIAL

25 February 2022 98/100 Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains **THREE (3)** short questions and **TEN (10)** MCQ questions.
2. Answer **ALL** questions.
3. This is an open book test.
4. All questions have ONLY ONE correct answer for the MCQ questions.

PART I – SHORT QUESTIONS

1. Calculate the entropy change when a piece of PVC is compressed from 1 to 5 atmospheric pressures at 298 K. The linear coefficient of thermal expansion of PVC is $60 \times 10^{-6}/°C$. The volume of the PVC is 1 m^3.

(10 marks)

$T = 298K$ $\alpha_l = 60\times10^{-6}\ °C^{-1} = 60\times10^{-6}\ K^{-1}$ $V = 1 m^3$

$\alpha_v = 3\alpha_l$. $P_1 = 1 atm$ $P_2 = 5 atm$.

$\because T$ is constant at 298K

$\left(\frac{\partial S}{\partial P}\right)_T = -V\alpha_v$

$= -3V\alpha_l$

where does this eqn come from?

$dS = -3V\alpha_l dP$

$\Delta S = \int_{P_1}^{P_2}(-3V\alpha_l)dP$

$= -3 \times 1m^3 \times 60\times10^{-6} K^{-1} \Big|_{1.013\times10^5 N/m^2}^{5\times1.013\times10^5 N/m^2}$

$= -72.936 J K^{-1}$

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2. Derive the relationship dF = – SdT – PdV, where F is the Helmholtz free energy, S is the entropy, V is the volume, P, the pressure and T, the temperature. Subsequently, prove that

$$-\left(\frac{dP}{dT}\right)_v = -\left(\frac{dS}{dV}\right)_T$$

(20 marks)

Assume in a isothermal condition.

the work done through reversible process:

$-\delta W = PdV$

~~$\delta W_{rev} = \ldots dU + (-PdV)$~~

~~$d = dU + PdV$~~

~~$dU = TdS - PdV$~~

~~$\delta W_{rev} =$~~

$\delta W_{rev} = dU - TdS$.

$\because T$ is constant,

$\therefore \delta W_{rev} = d(U - TS)$

$\int \delta W_{rev} = \int d U - TS$.

$W_{rev} = \Delta(U - TS)$ which defined as F.

$F \equiv U - TS$.

$dF = dU - TdS - SdT$.

$dU = TdS - PdV$.

$\therefore dF = - SdT - PdV$.

by Maxwell relation.

$F = F(T, V)$. $dF = (-S)\, dT + (-P) dV$

$-S = \left(\frac{\partial F}{\partial T}\right)_V \quad -P = \left(\frac{\partial F}{\partial V}\right)_T$

$\therefore \left(\frac{-\partial P}{\partial T}\right)_V = \left(-\frac{\partial S}{\partial V}\right)_T$. $-\left(\frac{\partial P}{\partial T}\right)_V = -\left(\frac{\partial S}{\partial V}\right)_T$

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3. (i) The pressure of an isolated system is reduced by an isochoric process. If the heat given out is 500 J in the process. Calculate the change in the internal energy of the system.

(ii) 1 mole of an ideal gas is compressed from a volume of $0.1\ m^3$ to $0.05\ m^3$ at a constant temperature of 100 K. Estimate the work done by ~~on~~ the gas in the process.

(20 marks)

i) ∵ is a isochoric process, $\delta W = PdV$

∴ There is no work done by gas.

$dU = \delta Q + \delta W$

$= \delta Q.$

$\int dU = \int \delta Q$

$\Delta U = \Delta Q = 500J.$

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ii)

$n = 1mol \quad V_1 = 0.1 m^3 \quad V_2 = 0.05 m^3. \quad T = 100K.$

∵ is a isothermal process, $\delta W = -PdV$

~~$\delta W = PdV$~~

~~$\int \delta W = \int PdV$~~

~~$\delta W = \int_{V_1}^{V_2} PdV$~~

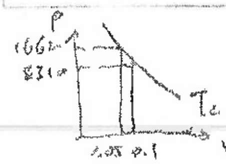
∵ Is an ideal gas

∴ $PV = nRT$, where nRT is a constant.

$P_1V_1 = nRT = P_2V_2$, $0.1P_1 = 0.05P_2$, $\frac{P_1}{P_2} = \frac{1}{2}$

$1 \times 8.31 \times 100 = 0.1 P_1 \quad P_1 = 8310 Pa.$

$P_2 = 16620 Pa.$ $\delta W = PdV = -\int_{0.05}^{0.1} P\, dV$



∵ The ΔV is very small, $-\int_{0.05}^{0.1} PdV$ can estimate by taking P's average. $\bar{P} = \frac{P_1+P_2}{2} = 12465 Pa.$

$\delta W = -623.25J$ (if no heat loss).

PART II MCQ QUESTIONS

(4) Answer all the **TEN MCQ** questions below. There is only **ONE** answer for each question.

(50 marks)

(a) If the pressure of the gas in a cylinder drops from 20 to 5 atmospheric pressures, calculate the final temperature of the gas if the initial temperature is 298 K.

monoatomic ideal gas

A 160 K

B 170 K

C 180 K

D 190 K

E 200 K

ANS: B

(b) Isentropic process is also a/an ____________?

A isothermal process

B isobaric process

C steady state process

D isotropic process

E adiabatic process

ANS: E

(c) Which of the following relationship is **NOT** correct?

A dU = TdS – PdV: $-\left(\frac{\partial P}{\partial S}\right)_V = \left(\frac{\partial T}{\partial V}\right)_S$

B dH = TdS + VdP: $\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$

C dF = -SdT – PdV: $-\left(\frac{\partial P}{\partial T}\right)_V = -\left(\frac{\partial S}{\partial V}\right)_T$

D dG = -SdT + VdP: $-\left(\frac{\partial S}{\partial T}\right)_P = \left(\frac{\partial V}{\partial P}\right)_T$

E dZ = NdX + MdY $\left(\frac{\partial N}{\partial Y}\right)_X = \left(\frac{\partial M}{\partial X}\right)_Y$

ANS: D

(d) Which of the following is NOT a condition for equilibrium?

(A) Free energy is minimum

(B) Entropy is maximum

(C) accessible microstate is minimum

(D) constant macrostate

(E) Enthalpy is maximum

ANS: D C

(e) Which of the following statement is TRUE?

A in an isothermal process heat may be given out or absorbed

B in an isothermal process, no heat is absorbed or given out

C in an adiabatic process, temperature is constant

D adiabatic process is always an isothermal process

E isothermal process is always also isobaric

ANS: A

(f) If a steel can and rubber can of the same shape are dropped together onto a concrete floor, which will come into thermodynamic equilibrium faster?

A Neither will come into equilibrium

B Steel can

C Both will reach equilibrium at the same time

D rubber can

E None of the above

ANS: B

(g) At constant pressure, the rate of change of the Gibbs free energy with respect to temperature is given by:

A entropy

B enthalpy

C heat capacity

D pressure

E volume

ANS: A or B

(h) "in all energy exchanges, if no energy enters or leaves the system, the potential energy of the state will always be less than that of the initial state." The statement is synonymous to:

A First Law of Thermodynamics

B Second Law of Thermodynamics

C Third Law of Thermodynamics

D Zeroth Law of Thermodynamics

E Gibbs Criteria for Equilibrium

ANS: B

(i) Which of the following properties is not an energy term?

A Pressure

B Enthalpy

C Entropy

D Heat

E Work

ANS: A

(j) A black box is sitting over a hole in a table. It is isolated in every way from its surroundings with the exception of a very thin thread which is connected to a weight. You observe the weight slowly moving upwards towards the box. Which of the following is true?

A This situation violates the First Law

B The First Law is satisfied, the energy in the box is decreasing

C The First Law is satisfied, the energy in the box is increasing

D The First Law is satisfied, the energy in the box is constant

E This situation violates the Second Law

ANS: B